

Inverse Problems in Geophysics

Exercise 4: Problem types & SVD

MGPY+MGIN+PhD

Thomas Günther, thomas.guenther@geophysik.tu-freiberg.de

Inverse problem types - classification scheme

$$\textcircled{!} \mathbf{d} = \mathbf{G}\mathbf{m} + \mathbf{n} \Rightarrow \mathbf{G} \in \mathcal{R}^{N \times M}$$

rows = data (N), columns = model parameter (M), rank r determines type of problem

 **Even-determined (LS=MN)**

$$M = N = r, \mathbf{R}^M = \mathbf{I}, \mathbf{R}^D = \mathbf{I}$$

 **Over-determined (LS)**

$$N > r = M, \mathbf{R}^M = \mathbf{I}, \mathbf{R}^D \neq \mathbf{I}$$

 **Under-determined (MN)**

$$N = r < M, \mathbf{R}^M \neq \mathbf{I}, \mathbf{R}^D = \mathbf{I}$$

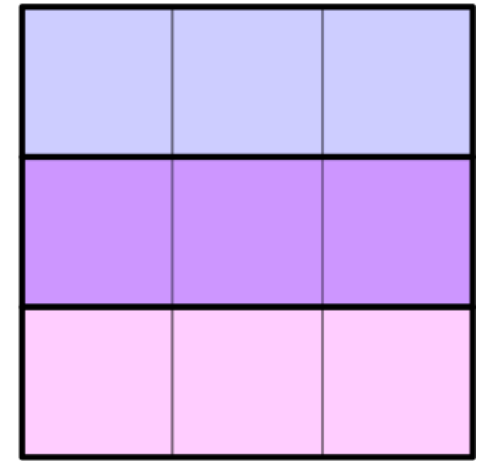
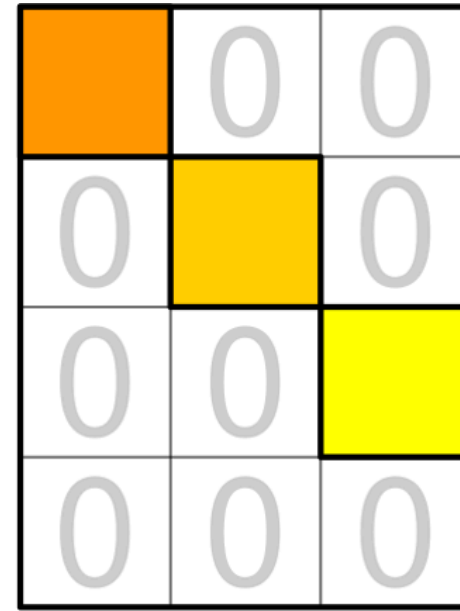
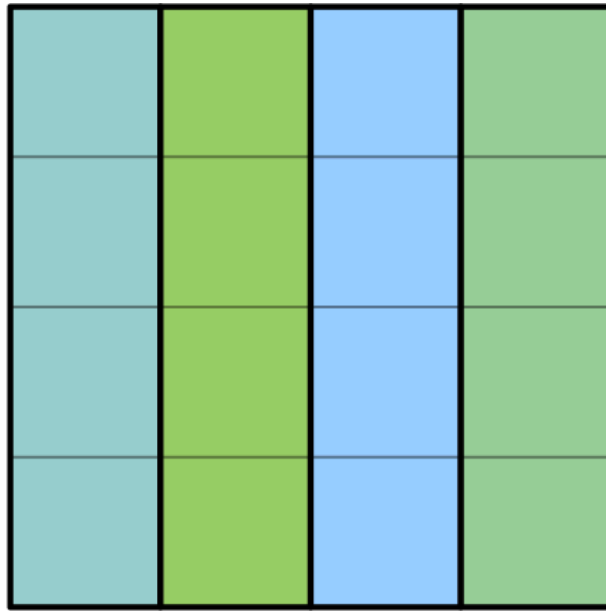
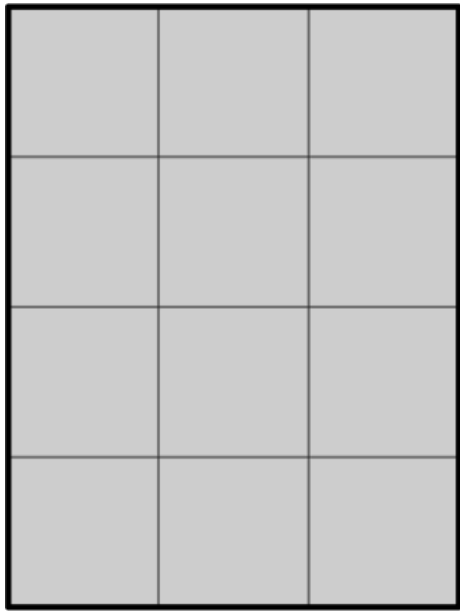
 **Mixed-determined**

$$r < N, r < M, \mathbf{R}^M \neq \mathbf{I}, \mathbf{R}^D \neq \mathbf{I}$$

 **SVD**

SVD provides the same solution for all

Singular value decomposition (SVD)



$$\mathbf{G} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^*$$

$$\mathbf{G} \in R^{N \times M} \quad \mathbf{U} \in R^{N \times N} \quad \mathbf{\Sigma} \in R^{N \times M} \quad \mathbf{V} \in R^{M \times M}$$

Problem types

Example linear problem: Traveltime tomography

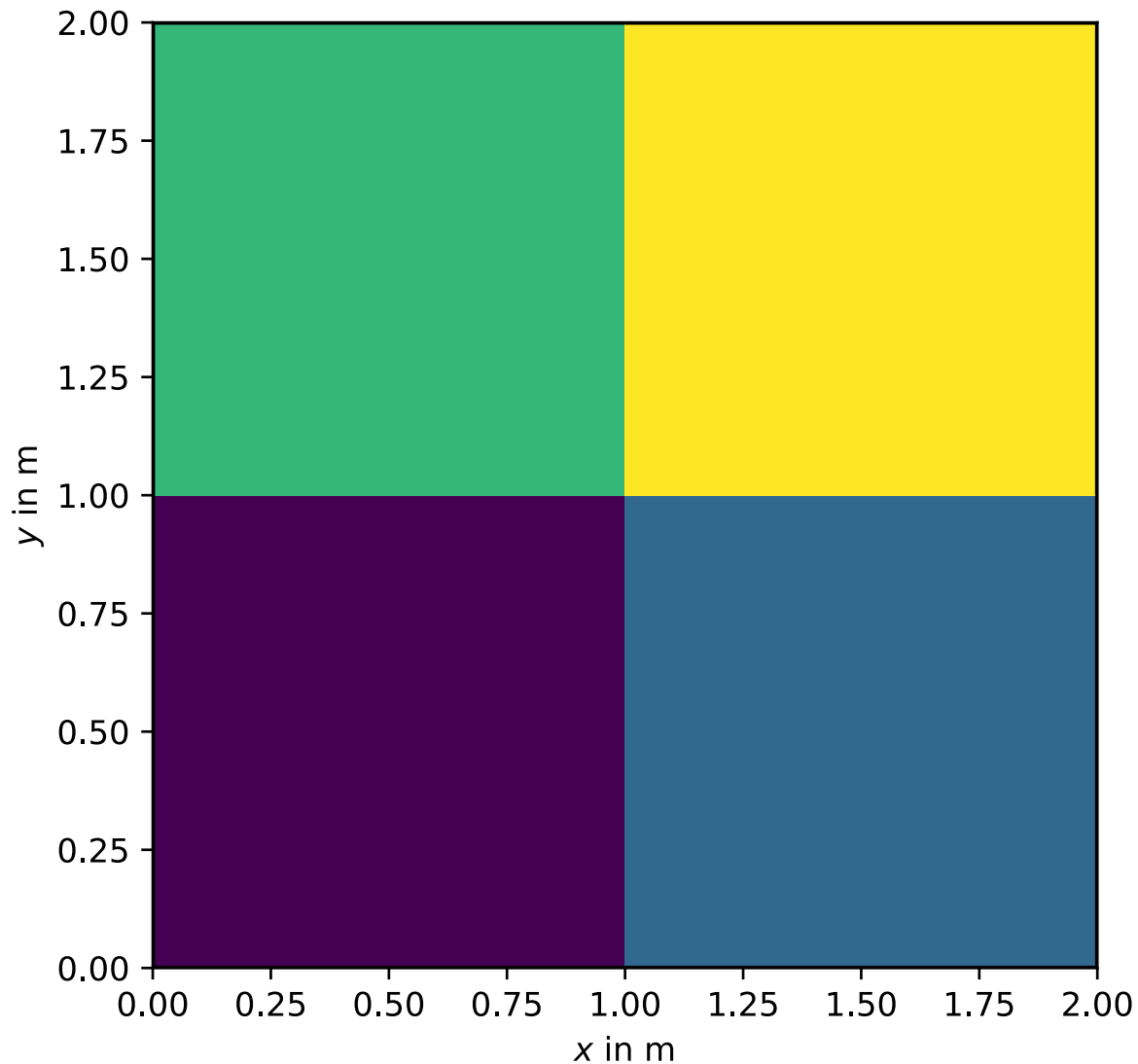
$$t = \int_L \frac{1}{v} dl = \int_L s dl$$

piece-wise constant s : $t = \sum_j s_j \Delta l_j \Rightarrow t_i = \sum W_{ij} s_j$

$\mathbf{d} = \mathbf{W}\mathbf{m}$ where $\mathbf{W}$ contains lengths of rays (rows) in cells (columns)

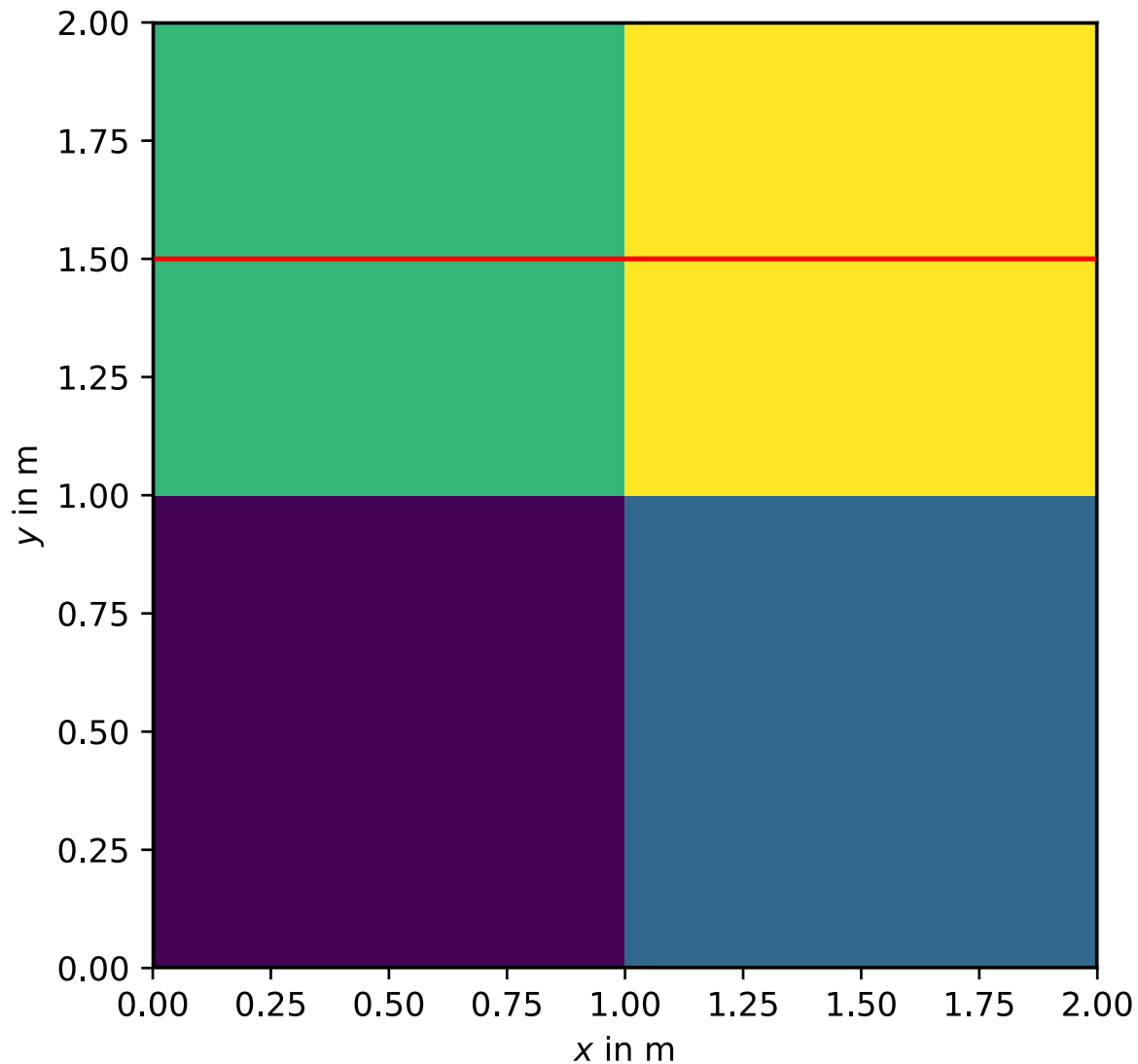
with $\mathbf{d} = [t_1, \dots, t_N]$, $\mathbf{m} = [s_1, \dots, s_M]$

Problem types - model



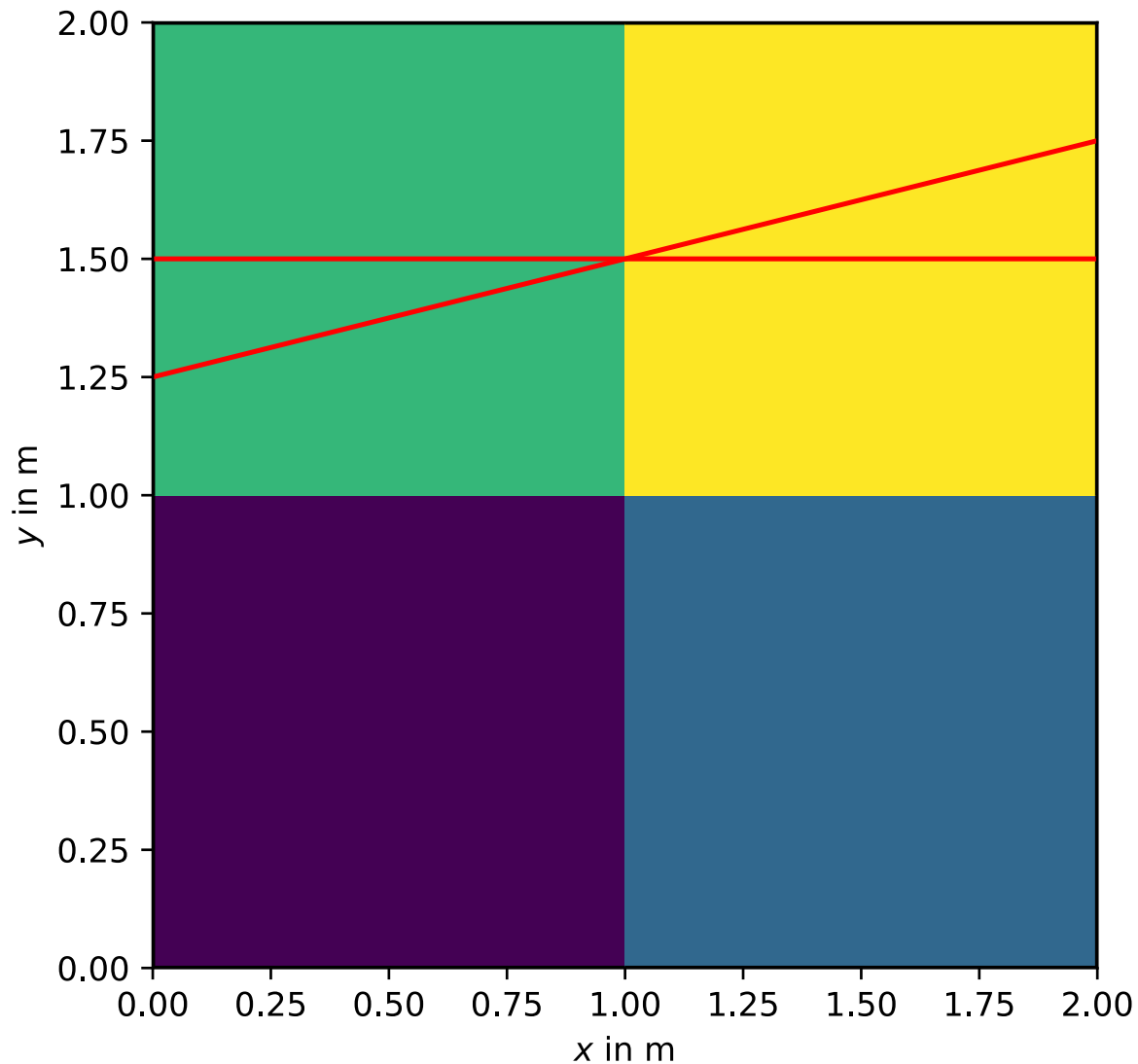
- guess whether it is a) over-, b) under-, or c) mixed-determined
- generate the ray path matrix
- determine the rank by using SVD
- generate data from an inhomogeneous model vector
- choose method & compute model
- compare with SVD solution (`pinv`)
- compile the numbers in a table

Problem types - 1



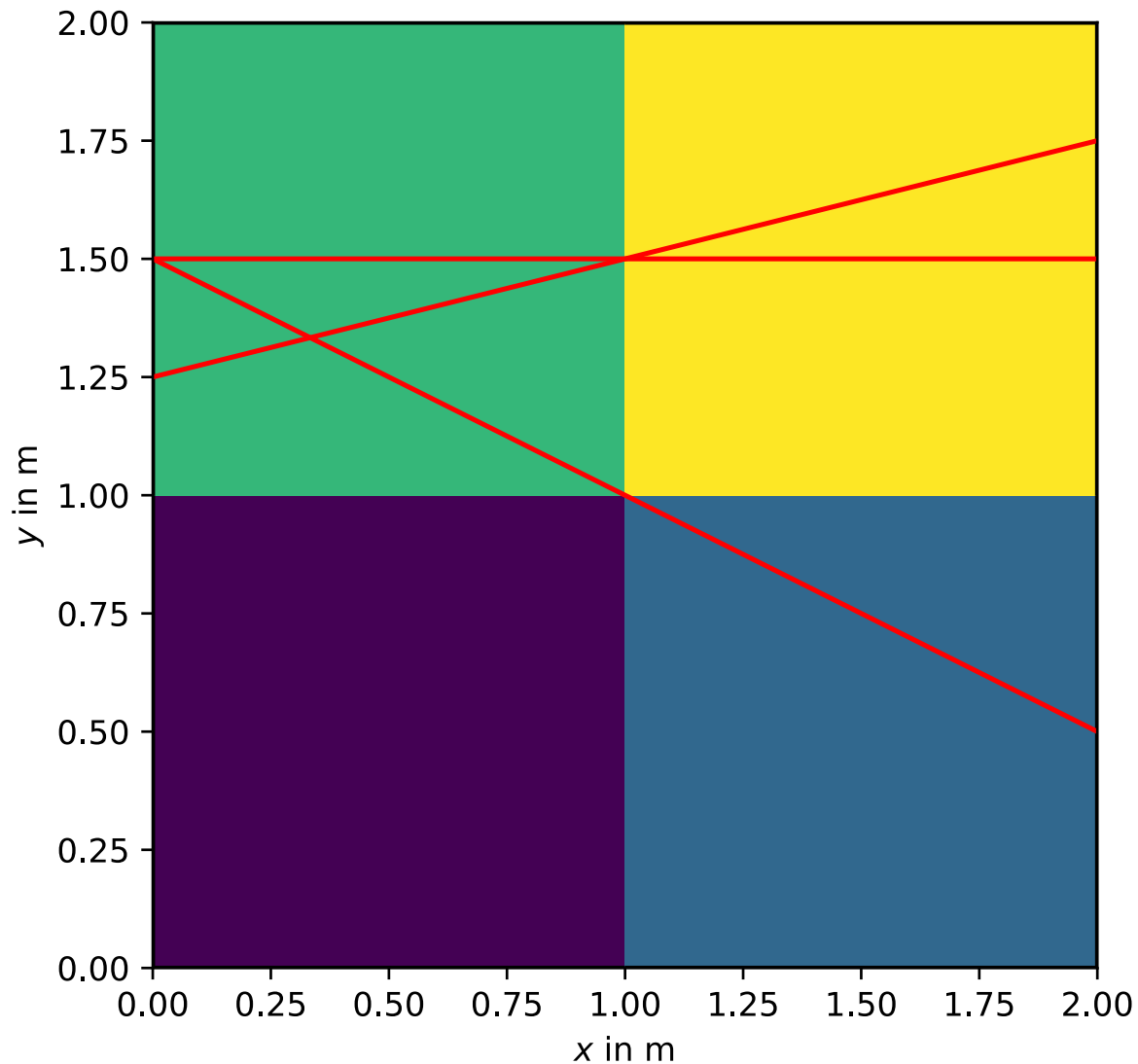
- guess whether it is a) over-, b) under-, or c) mixed-determined
- generate the ray path matrix
- determine the rank by using SVD
- generate data from an inhomogeneous model vector
- choose method & compute model
- compare with SVD solution (`pinv`)
- compile the numbers in a table

Problem types - 2



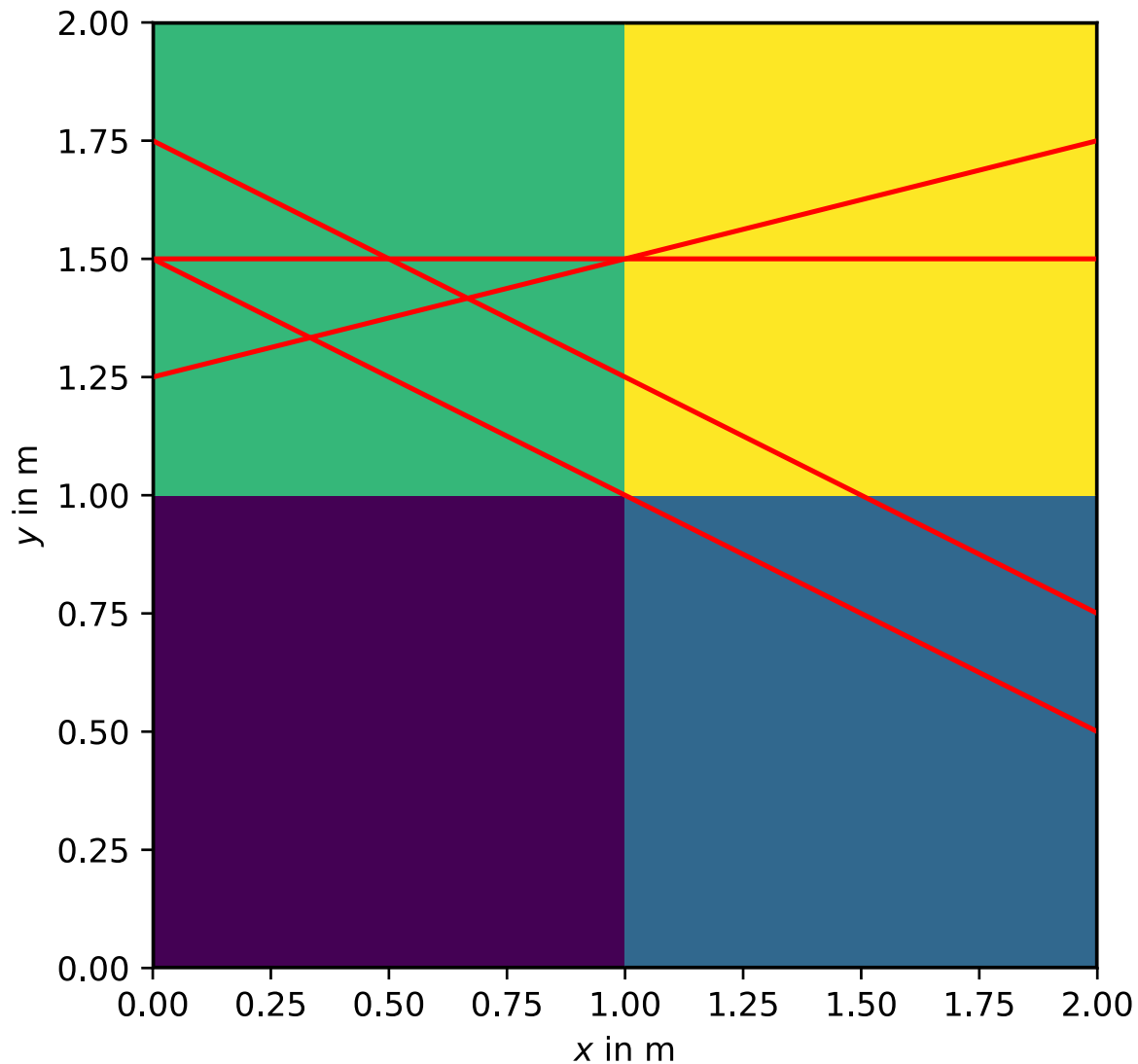
- guess whether it is a) over-, b) under-, or c) mixed-determined
- generate the ray path matrix
- determine the rank by using SVD
- generate data from an inhomogeneous model vector
- choose method & compute model
- compare with SVD solution (`pinv`)
- compile the numbers in a table

Problem types - 3



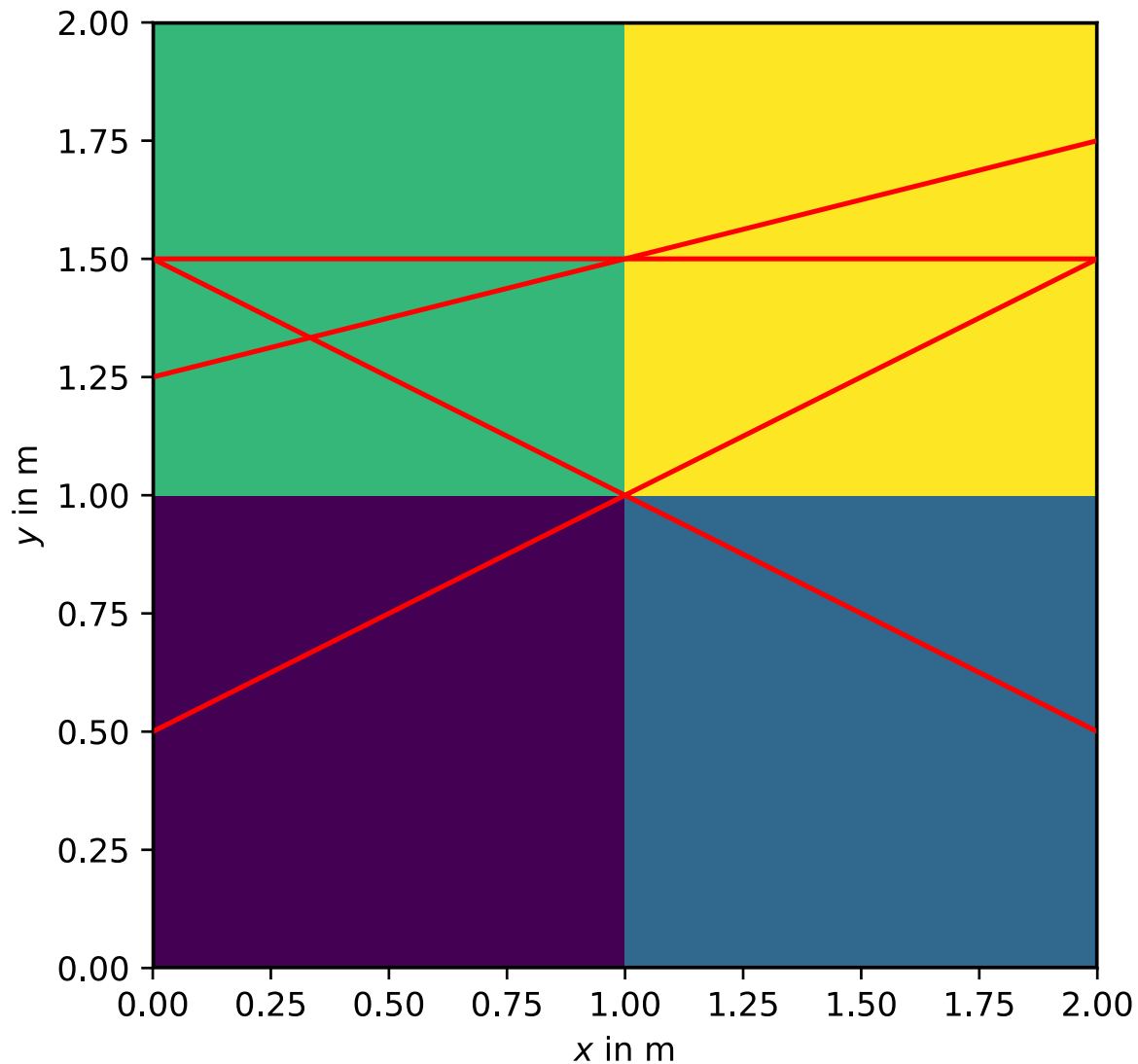
- guess whether it is a) over-, b) under-, or c) mixed-determined
- generate the ray path matrix
- determine the rank by using SVD
- generate data from an inhomogeneous model vector
- choose method & compute model
- compare with SVD solution (`pinv`)
- compile the numbers in a table

Problem types - 4



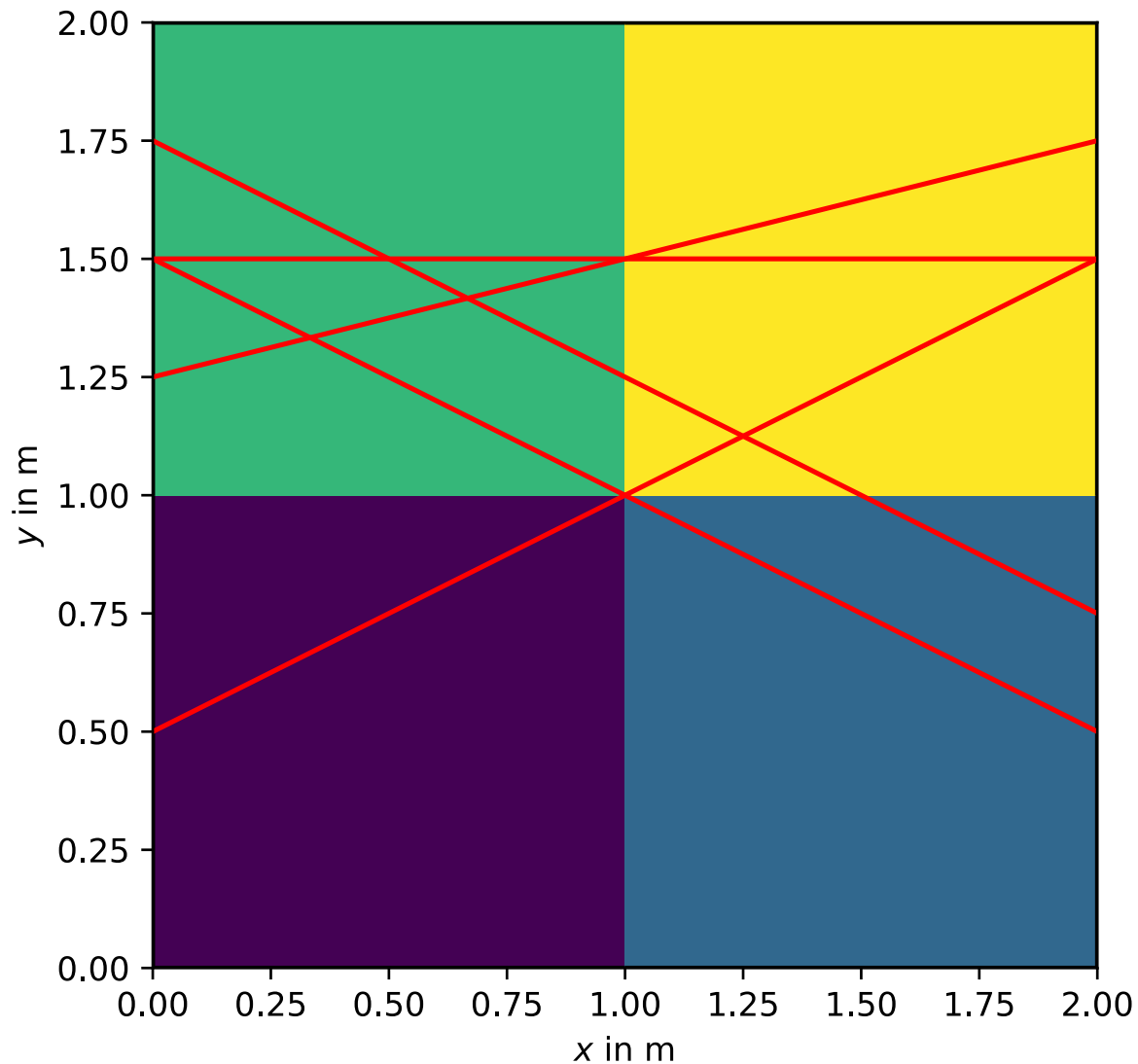
- guess whether it is a) over-, b) under-, or c) mixed-determined
- generate the ray path matrix
- determine the rank by using SVD
- generate data from an inhomogeneous model vector
- choose method & compute model
- compare with SVD solution (`pinv`)
- compile the numbers in a table

Problem types - 5



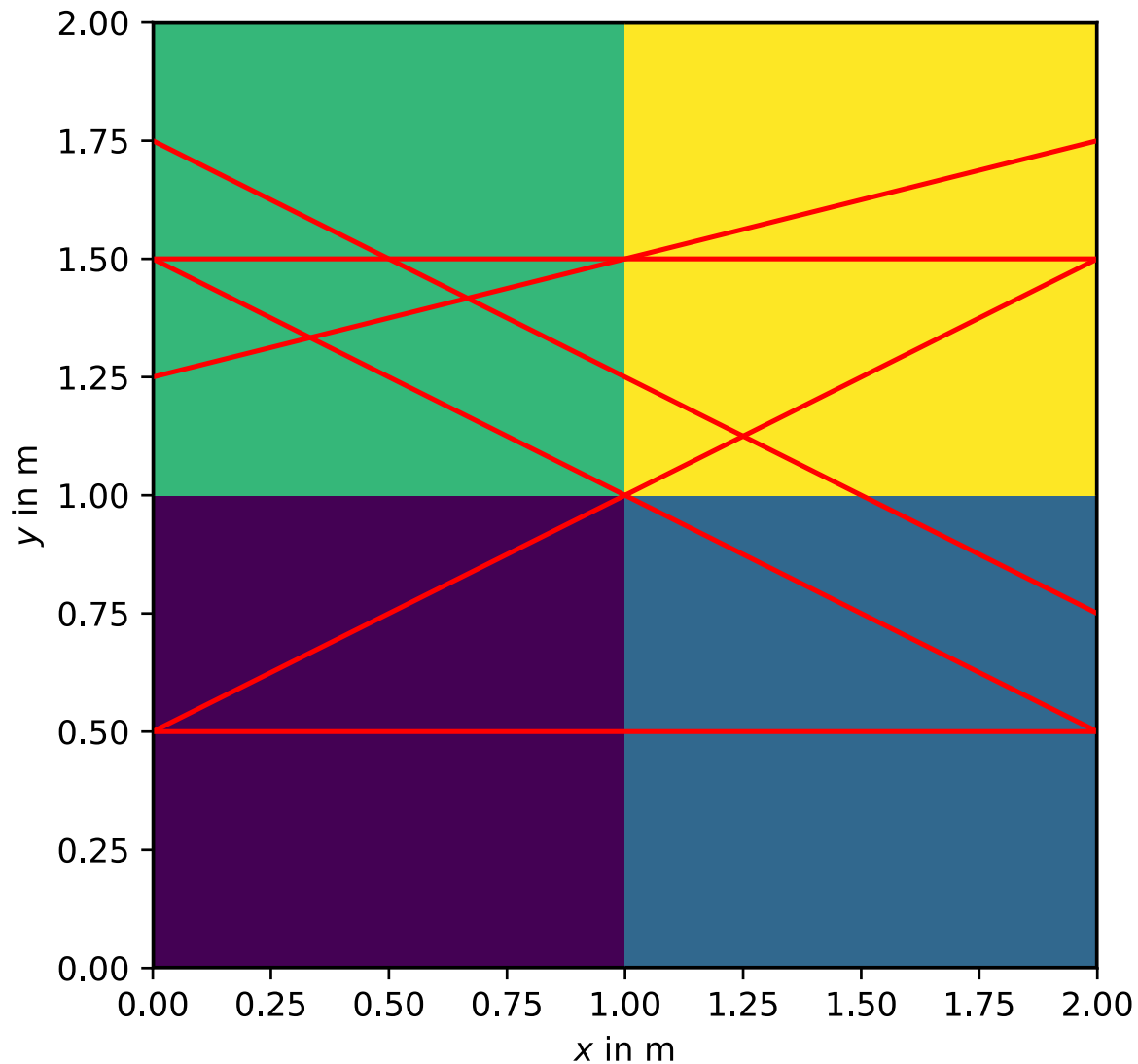
- guess whether it is a) over-, b) under-, or c) mixed-determined
- generate the ray path matrix
- determine the rank by using SVD
- generate data from an inhomogeneous model vector
- choose method & compute model
- compare with SVD solution (`pinv`)
- compile the numbers in a table

Problem types - 6



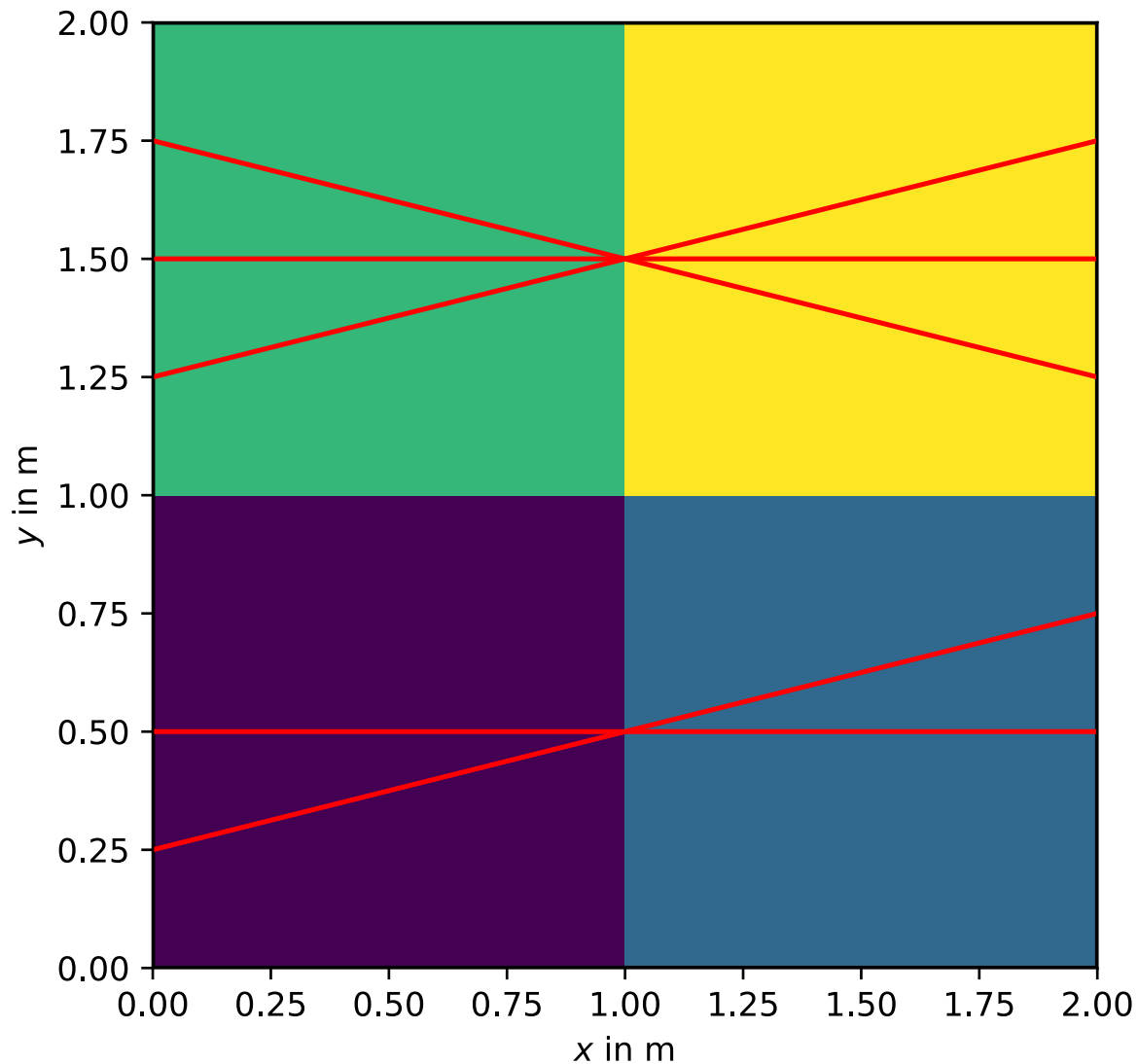
- guess whether it is a) over-, b) under-, or c) mixed-determined
- generate the ray path matrix
- determine the rank by using SVD
- generate data from an inhomogeneous model vector
- choose method & compute model
- compare with SVD solution (`pinv`)
- compile the numbers in a table

Problem types - 7



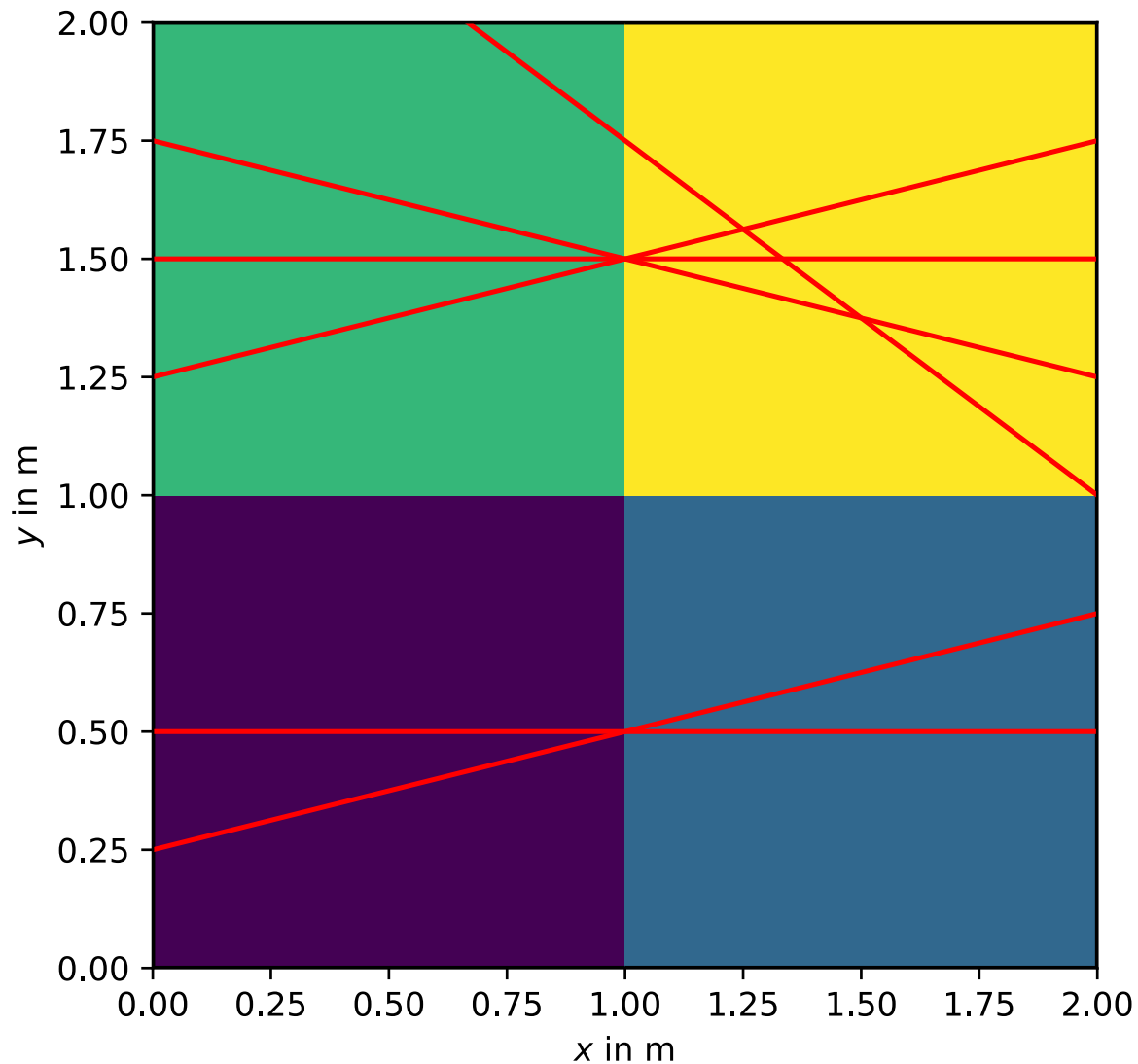
- guess whether it is a) over-, b) under-, or c) mixed-determined
- generate the ray path matrix
- determine the rank by using SVD
- generate data from an inhomogeneous model vector
- choose method & compute model
- compare with SVD solution (`pinv`)
- compile the numbers in a table

Problem types - 8



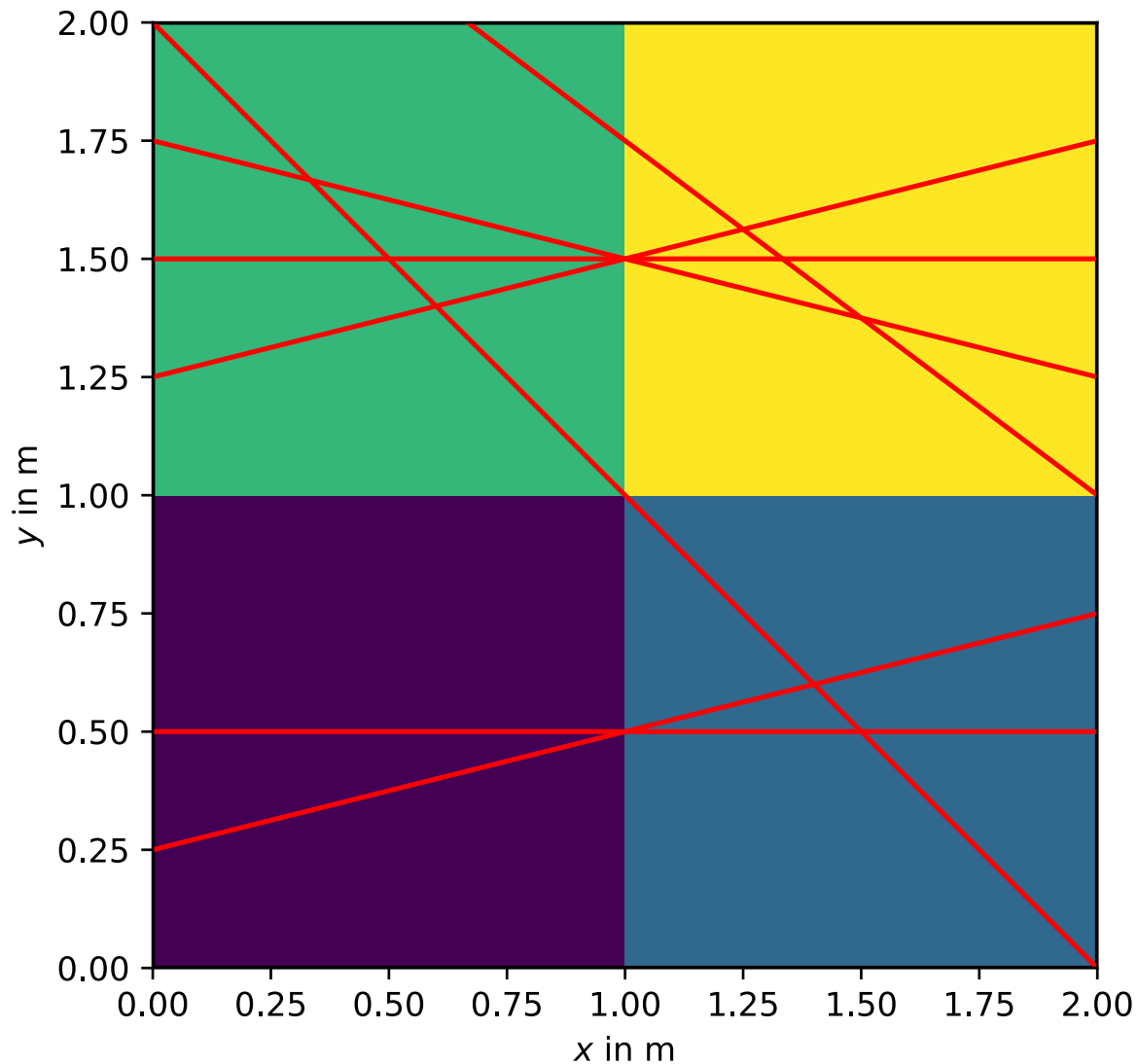
- guess whether it is a) over-, b) under-, or c) mixed-determined
- generate the ray path matrix
- determine the rank by using SVD
- generate data from an inhomogeneous model vector
- choose method & compute model
- compare with SVD solution (`pinv`)
- compile the numbers in a table

Problem types - 9



- guess whether it is a) over-, b) under-, or c) mixed-determined
- generate the ray path matrix
- determine the rank by using SVD
- generate data from an inhomogeneous model vector
- choose method & compute model
- compare with SVD solution (`pinv`)
- compile the numbers in a table

Problem types - 10



- guess a) even, b) over-, c) under-, or d) mixed-determined
- generate the ray path matrix
- determine the rank by using SVD
- generate data from an inhomogeneous model vector
- choose method & compute model
- compare with SVD solution (`pinv`)
- compile the numbers in a table

Additionally use SVD

- compute SVD of the system
- have a look at eigenvalues and eigenvectors
- compute model and data resolution matrices
- compute solution and compare with LS and MN solutions